

# Scaling Book Exercises: Section 6

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**Chapter:** Training LLaMA 3 on TPUs

**Source scan:** ../handwritten/section\_06\_handwritten.pdf

**Reference:** <https://jax-ml.github.io/scaling-book/applied-training/>

**Hardware and training assumptions.** A full TPU v5p pod has

$$16 \times 20 \times 28 = 8960 \text{ chips.}$$

Each chip has bfloat16 peak compute

$$C = 4.59 \times 10^{14} \text{ FLOPs/s.}$$

The ICI operational intensity is approximately 2550, and the DCN operational intensity is

$$\frac{C}{W_{\text{DCN}}} = \frac{4.59 \times 10^{14}}{6.25 \times 10^9} \approx 73,440.$$

The batch size is  $B = 2^{22} = 4,194,304$  tokens and MFU is 40%.

## Question 1: Scaling LLaMA 3-70B to four pods

Use data parallelism across the four pods and mixed FSDP plus tensor parallelism within each pod. Each pod receives

$$B_{\text{pod}} = \frac{4,194,304}{4} = 1,048,576 \text{ tokens.}$$

This is well above the DCN requirement:

$$1,048,576 > 73,440,$$

so inter-pod data-parallel communication can be hidden by compute.

Across all four pods, the local batch is

$$\frac{B}{4(8960)} \approx 117.0 \text{ tokens per chip.}$$

For LLaMA 3-70B,  $F = 28,672$ , so the mixed FSDP and tensor-parallel threshold is

$$\frac{2550^2}{2F} = \frac{2550^2}{2(28672)} \approx 113.4 \text{ tokens per chip.}$$

Since  $117.0 > 113.4$ , the continuous roofline model says this setup is barely compute-bound.

Within each pod, the idealized FSDP degree is

$$\begin{aligned} X_{\text{opt}} &= \sqrt{\frac{2B_{\text{pod}}N_{\text{pod}}}{F}} \\ &= \sqrt{\frac{2(1,048,576)(8960)}{28,672}} \\ &\approx 810, \end{aligned}$$

with  $Y = N_{\text{pod}}/X \approx 11$ -way tensor parallelism. The exact integer layout would be rounded to match the physical pod topology.

The one-pod estimate is approximately 44 days at 40% MFU. Four pods provide four times the compute, so

$$T_{\text{train}} \approx \frac{44}{4} = 11 \text{ days}.$$

## Question 2: LLaMA 3-405B

### (a) Parameters and training FLOPs

The model configuration is

| Hyperparameter             | Value   |
|----------------------------|---------|
| Layers $L$                 | 126     |
| Model dimension $D$        | 16,384  |
| Feed-forward dimension $F$ | 53,248  |
| Query heads $N$            | 128     |
| KV heads $K$               | 8       |
| Head dimension $H$         | 128     |
| Vocabulary size $V$        | 128,256 |

The gated MLP contributes

$$P_{\text{MLP}} = 3LDF = 329,772,957,696.$$

Grouped-query attention contributes

$$\begin{aligned} P_{\text{attention}} &= 2LDH(N + K) \\ &= 71,873,593,344. \end{aligned}$$

The separate input and output embeddings contribute

$$P_{\text{vocab}} = 2VD = 4,202,692,608.$$

Therefore,

$$\begin{aligned} P &= P_{\text{MLP}} + P_{\text{attention}} + P_{\text{vocab}} \\ &= 405,849,243,648 \\ &\approx \boxed{405.85\text{B parameters}}. \end{aligned}$$

Training takes approximately six FLOPs per parameter per token:

$$\boxed{\text{FLOPs/token} \approx 6P = 2.435 \times 10^{12}}.$$

For 15 trillion tokens,

$$\begin{aligned} F_{\text{total}} &= 6P(15 \times 10^{12}) \\ &= 3.653 \times 10^{25} \text{ FLOPs.} \end{aligned}$$

## (b) Training on eight pods

Use data parallelism across the eight pods, with mixed FSDP and tensor parallelism inside each pod. Each pod receives

$$B_{\text{pod}} = \frac{4,194,304}{8} = 524,288 \text{ tokens.}$$

Since  $524,288 > 73,440$ , the inter-pod DCN AllReduce is below its roofline bound.

The local batch across all  $8(8960) = 71,680$  chips is

$$\frac{B}{71,680} \approx 58.5 \text{ tokens per chip.}$$

For  $F = 53,248$ , the mixed-parallel compute-bound threshold is

$$\frac{2550^2}{2F} = \frac{2550^2}{2(53248)} \approx 61.1 \text{ tokens per chip.}$$

Thus the configuration is slightly communication-bound within each pod. Pure FSDP would be much further below its approximately 850-token local batch requirement.

The continuous optimum inside one pod is

$$X_{\text{opt}} = \sqrt{\frac{2B_{\text{pod}}N_{\text{pod}}}{F}} \approx 420, \quad Y_{\text{opt}} \approx 21.$$

A nearby topology-aligned split is  $X = 448$  FSDP shards and  $Y = 20$  tensor shards.

Ignoring communication gives the compute lower bound

$$\begin{aligned} T_{\text{compute}} &= \frac{3.653 \times 10^{25}}{8(8960)(0.4)(4.59 \times 10^{14})} \\ &= 2.775 \times 10^6 \text{ s} \\ &\approx 32.1 \text{ days.} \end{aligned}$$

Because the ideal mixed layout is only slightly communication-bound, the roofline estimate rises to approximately

$$\boxed{32\text{--}33 \text{ days}}.$$